

DATA ANALYSIS & INTERPRETATION



SIT225

DATA ANALYSIS

- Data Analysis is the technique of collecting, transforming, and organizing **data** to **make future predictions** and **informed data-driven decisions**.
- It generally starts with asking a question, it follows a data collection process which will **be examined to answer the asked question**.

ANALYSIS STEPS

- Given the existence of data, analysis steps commonly consists of
 - Clear and process
 - Analyse
 - Share and
 - Act or report

The Data Analysis Process



DATA STATISTICS

- Statistics is the science of analyzing data
- Basic descriptive statistics, which includes
Count, sum, standard deviation, average, percentile and correlation

	Duration	Average_Pulse	Max_Pulse	Calorie_Burnage	Hours_Work	Hours_Sleep
count	163.0	163.0	163.0	163.0	163.0	163.0
mean	64.26	107.72	134.23	382.37	4.39	7.68
std	42.99	14.63	16.4	274.23	3.92	0.66
min	15.0	80.0	100.0	50.0	0.0	5.0
25%	45.0	100.0	124.0	256.5	0.0	7.5
50%	60.0	105.0	131.0	320.0	5.0	8.0
75%	60.0	111.0	141.0	388.5	8.0	8.0
max	300.0	159.0	184.0	1860.0	11.0	12.0

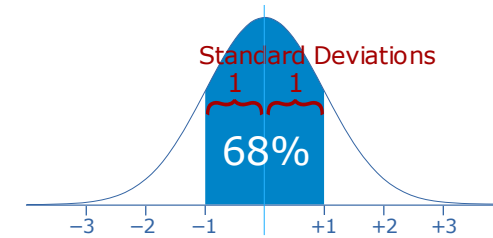
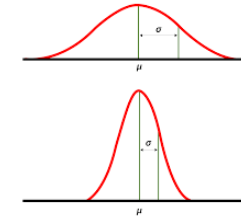
PERCENTILE

	Duration	Average_Pulse	Max_Pulse	Calorie_Burnage	Hours_Work	Hours_Sleep
count	163.0	163.0	163.0	163.0	163.0	163.0
mean	64.26	107.72	134.23	382.37	4.39	7.68
std	42.99	14.63	16.4	274.23	3.92	0.66
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- A number that describes the **value that a given percent of the values are lower than**.
- The 25% percentile of Average_Pulse means that 25% of all the training sessions have an average pulse of 100 beats per minute or lower.
- If we flip the statement, it means that 75% of all the training sessions have an average pulse of 100 beats per minute or higher.
- The 75% percentile of Average_Pulse means that 75% of all the training session have an average pulse of 111 or lower.

STD DEVIATION

- A standard deviation (or σ) is a measure of how dispersed the data is in relation to the mean.
- Low, or small, standard deviation indicates data are clustered tightly around the mean, and high, or large, standard deviation indicates data are more spread out.



$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

```
import numpy as np

std = np.std(full_health_data)
print(std)
```

VARIANCE

```
import numpy as np

var = np.var(health_data)
print(var)
```

- Variance is the average squared deviations from the mean, while standard deviation is the square root of this number.
- Both measures reflect variability in a distribution, but their units differ:
Standard deviation is expressed in the same units as the original values (e.g., minutes or meters).
Variance is expressed in much larger units (e.g., meters squared).
- Although the units of variance are harder to intuitively understand, variance is important in **statistical tests**.

STATISTICAL TEST

- Statistical tests are used in hypothesis testing.
- Hypothesis testing is a formal procedure for investigating our ideas about the world using statistics.
- It is most often used by scientists to test specific predictions, called hypotheses, that arise from theories.
- There are 5 main steps in hypothesis testing:

State your research hypothesis as a null hypothesis and alternate hypothesis (H_0) and (H_a or H_1)

STATISTICAL TEST

- There are 5 main steps in hypothesis testing:
 - ❑ State your research hypothesis as a null hypothesis and alternate hypothesis (H_0) and (H_a or H_1)
 - ❑ Collect data in a way designed to test the hypothesis.
 - ❑ Perform an appropriate statistical test.
 - ❑ Decide whether to reject or fail to reject your null hypothesis.
 - ❑ Present the findings in your results and discussion section.

<https://www.scribbr.com/statistics/hypothesis-testing/>

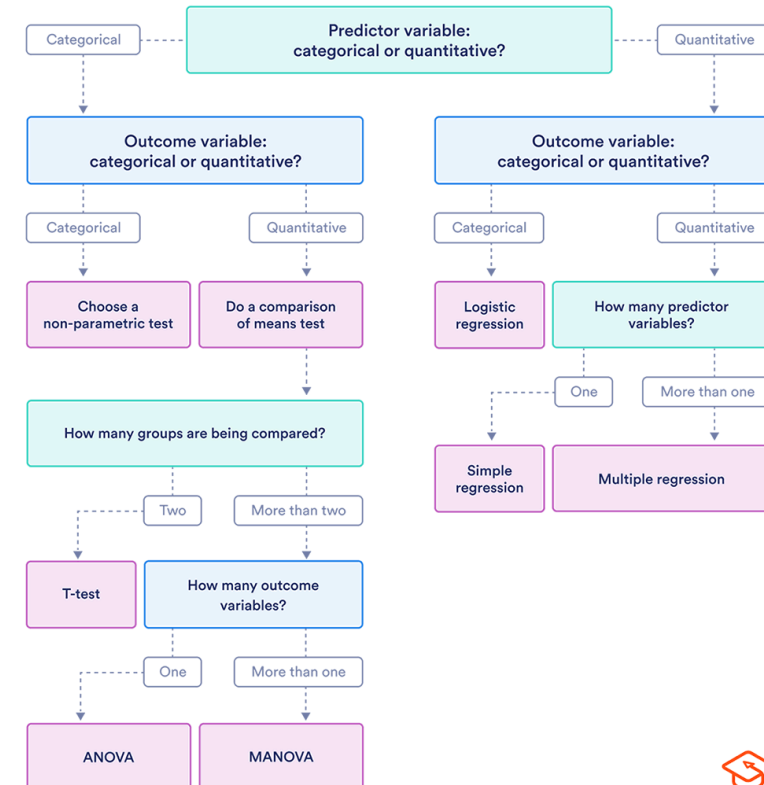
CHOOSING THE RIGHT STATISTICAL TEST

- Statistical tests can be used to:
 - ❑ determine whether a predictor variable has a statistically significant relationship with an outcome variable.
 - ❑ estimate the difference between two or more groups.
- Statistical tests assume a null hypothesis of no relationship or no difference between groups.
- Then they determine whether the observed data fall outside of the range of values predicted by the null hypothesis.

CHOOSING A STATISTICAL TEST

Choosing a statistical test

This flowchart helps you choose among parametric tests

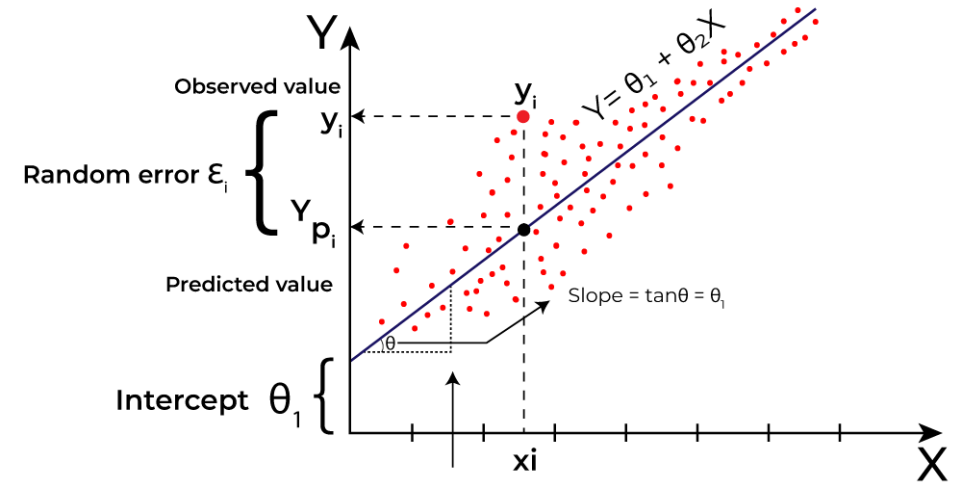


CORRELATION

- Correlation measures the relationship between two variables.
- The correlation coefficient measures the relationship between two variables.
- The correlation coefficient can never be less than -1 or higher than 1.
 - ❑ 1 = there is a perfect linear relationship between the variables.
 - ❑ 0 = there is no linear relationship between the variables.
 - ❑ -1 = there is a perfect negative linear relationship between the variables.

LINEAR REGRESSION

- Linear Regression is a machine learning algorithm based on supervised learning.
- It performs a regression task.
- Regression models a target prediction value based on independent variables.
- It is mostly used for finding out the relationship between variables and forecasting.
- where X is the explanatory variable and Y is the dependent variable.
- The slope of the line is b , and a is the intercept (the value of y when $x = 0$)

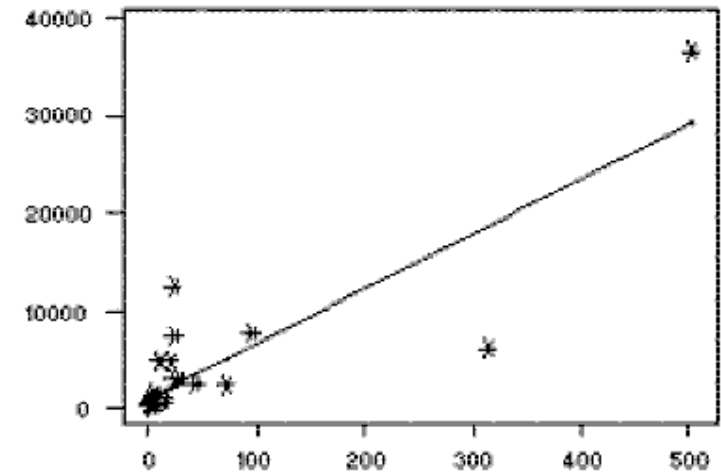


EXAMPLE

- The dataset "Televisions, Physicians, and Life Expectancy" contains, among other variables, the number of people per television set and the number of people per physician for 40 countries.
- Since both variables probably reflect the level of wealth in each country, it is reasonable to assume that there is some positive association between them.
- After removing 8 countries with missing values from the dataset, the remaining 32 countries have a correlation coefficient of 0.852 for number of people per television set and number of people per physician.
- The r^2 value is 0.726 (the square of the correlation coefficient), indicating that 72.6% of the variation in one variable may be explained by the other.
- Suppose we choose to consider number of people per television set as the explanatory variable, and number of people per physician as the dependent variable.

EXAMPLE...

- The regression equation is $\text{People.Phys.} = 1019 + 56.2 \text{ People.Tel.}$
- To view the fit of the model to the observed data, one may plot the computed regression line over the actual data points to evaluate the results.
- While most of the data points are clustered towards the lower left corner of the plot (indicating relatively few individuals per television set and per physician), there are a few points which lie far away from the main cluster of the data.
- These points are known as outliers, and depending on their location may have a major impact on the regression line



REGRESSION WITH PYTHON

- Number of Instances: 442
- Number of Attributes; First 10 columns are numeric predictive values
- Target: Column 11 is a quantitative measure of disease progression one year after baseline
- Attribute Information: ['age', 'sex', 'bmi', 'bp', 's1', 's2', 's3', 's4', 's5', 's6']

```
from sklearn import datasets, linear_model
from sklearn.metrics import mean_squared_error, r2_score

# Load the diabetes dataset
diabetes_X, diabetes_y = datasets.load_diabetes(return_X_y=True)

# Use only one feature
diabetes_X = diabetes_X[:, np.newaxis, 2]

# Split the data into training/testing sets
diabetes_X_train = diabetes_X[:-20]
diabetes_X_test = diabetes_X[-20:]

# Split the targets into training/testing sets
diabetes_y_train = diabetes_y[:-20]
diabetes_y_test = diabetes_y[-20:]
```

```
# Create linear regression object
regr = linear_model.LinearRegression()

# Train the model using the training sets
regr.fit(diabetes_X_train, diabetes_y_train)

# Make predictions using the testing set
diabetes_y_pred = regr.predict(diabetes_X_test)

# The coefficients
print("Coefficients: \n", regr.coef_)
```

```
# Plot outputs
plt.scatter(diabetes_X_test, diabetes_y_test, color="black")
plt.plot(diabetes_X_test, diabetes_y_pred, color="blue", linewidth=3)
```

REGRESSION WITH PYTHON

- Number of Instances: 442

- Num

num

- Targ

disea

- Attribute Information: Fasting blood sugar (0 to 125)

's2',

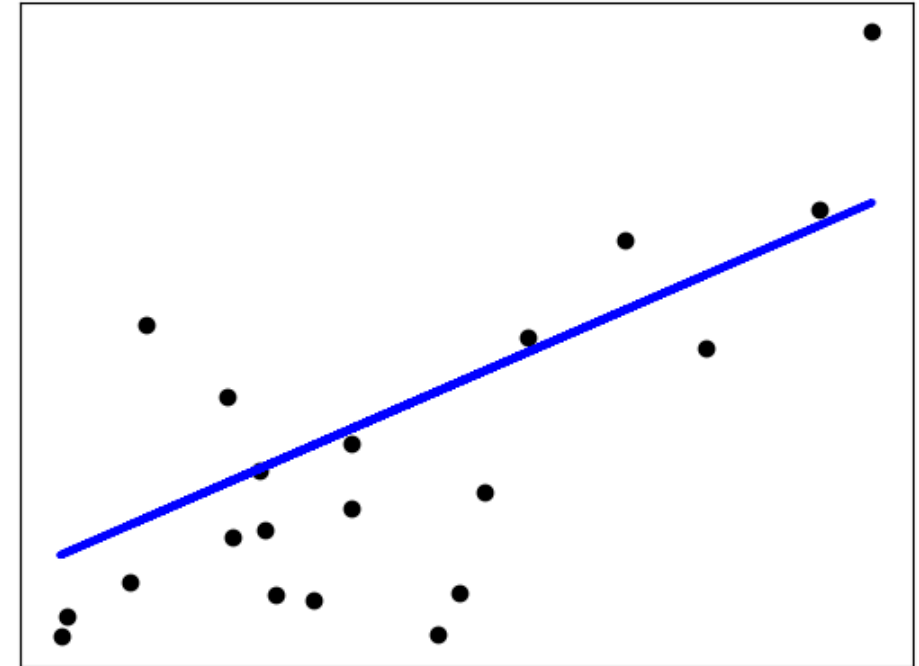
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Example shown in notebook.

LOGISTIC REGRESSION (CLASSIFICATION)

- Logistic regression is a statistical method for predicting binary classes.
- It is a special case of linear regression where the target variable is categorical in nature. It uses a **log of odds** as the dependent variable.
- Logistic Regression predicts the probability of occurrence of a binary event utilizing a logit function.

<https://www.ibm.com/topics/logistic-regression>

<https://www.datacamp.com/tutorial/understanding-logistic-regression-python>

LOGISTIC REGRESSION (CLASSIFICATION)...

- Linear Regression Equation,
Where y is a dependent variable and x1, x2 ... and Xn are explanatory variables.
- Sigmoid Function
- Apply Sigmoid function on linear regression.

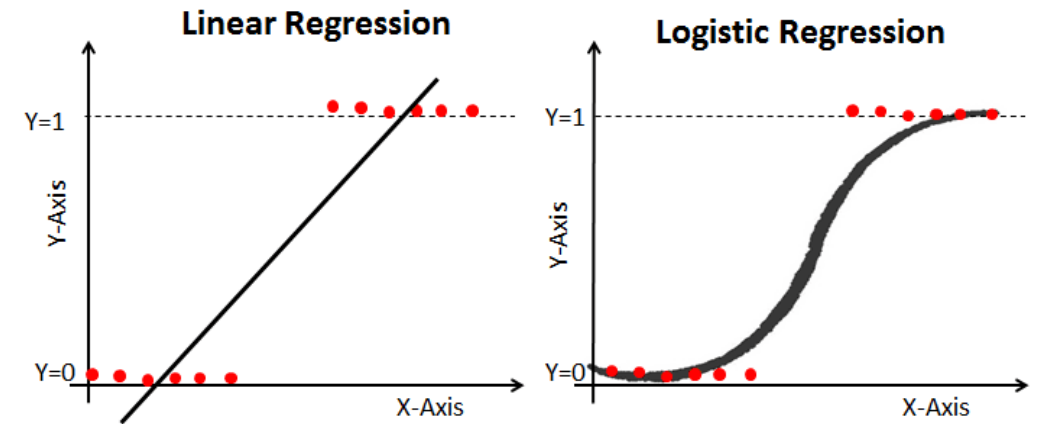
$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

$$p = \frac{1}{1 + e^{-y}}$$

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

LINEAR VS LOGISTIC REGRESSION

-
- Linear regression gives you a continuous output, but logistic regression provides a constant output.
 - An example of the continuous output is house price and stock price.
 - Examples of the discrete output are predicting whether a patient has cancer or not and predicting whether a customer will churn.
 - Logistic regression is estimated using the maximum likelihood estimation (MLE) approach, while linear regression is typically estimated using ordinary least squares (OLS).



LOGISTIC REGRESSION MODEL BUILDING

- Python sklearn package has LogisticRegression class
- Same approach of LinearRegression()
- Split data samples into train and test splits
- Train the model with training samples
- Test the model's performance from test samples.
- Performance metric include confusion matrix which counts (in)correct predictions w.r.t. ground truth.

```
# import the class
from sklearn.linear_model import LogisticRegression

# instantiate the model (using the default parameters)
logreg = LogisticRegression(random_state=16)

# fit the model with data
logreg.fit(X_train, y_train)

y_pred = logreg.predict(X_test)
```

```
# import the metrics class
from sklearn import metrics

cnf_matrix = metrics.confusion_matrix(y_test, y_pred)
cnf_matrix
```

<https://www.ibm.com/topics/logistic-regression>

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LOGISTIC REGRESSION MODEL BUILDING...

- 2-class classification problem, see notebook using pima diabetes dataset.
- 3-class classification problem, see link (https://scikit-learn.org/stable/auto_examples/linear_model/plot_iris_logistic.html)
- **D or HD students:** You can use your accelerometer activity data where you have labeled activities to formulate a 2- or 3-class activity classification problem (Task 8.3D).

<https://www.ibm.com/topics/logistic-regression>

<https://www.datacamp.com/tutorial/understanding-logistic-regression-python>

RECAP & FUTURE STEPS

- You have learned how to train a linear regression or logistic regression model using training data samples.
- To generate predictions (continuous output in regression, discrete output in logistic regression) from unseen test data.
- You should be data hungry at this point to harness your regression or classification skills to solve some real-world problems.

ANY QUESTION?

